

“CAIRO REAL ESTATE MARKET ANALYSIS”

SUBMITTED BY DATA DECODERS TEAM :

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INTRODUCTION AND WHAT WE FOUND

For this project, we decided to look at something relevant to everyday life in Cairo property prices. Housing is a big deal, especially with how fast the city is growing, and we were curious whether things like location or property type have a measurable impact on what people end up paying.

To explore this, we worked with a dataset of 11,418 property listings scraped from Egyptian real estate platforms. After cleaning it up, we ended up with 8,851 usable records spread across 28 districts and 10 different property types.

The short answer to what we found: yes, both location and property type matter a lot. Certain districts like El Salam, Zamalek and Garden City have prices per square meter that are over 50% above the city average. Villas and Twin Houses cost more than twice what a typical apartment. And we ran a proper statistical test (one-way ANOVA) to confirm these aren't just random patterns they're statistically significant.

DATA DESCRIPTION

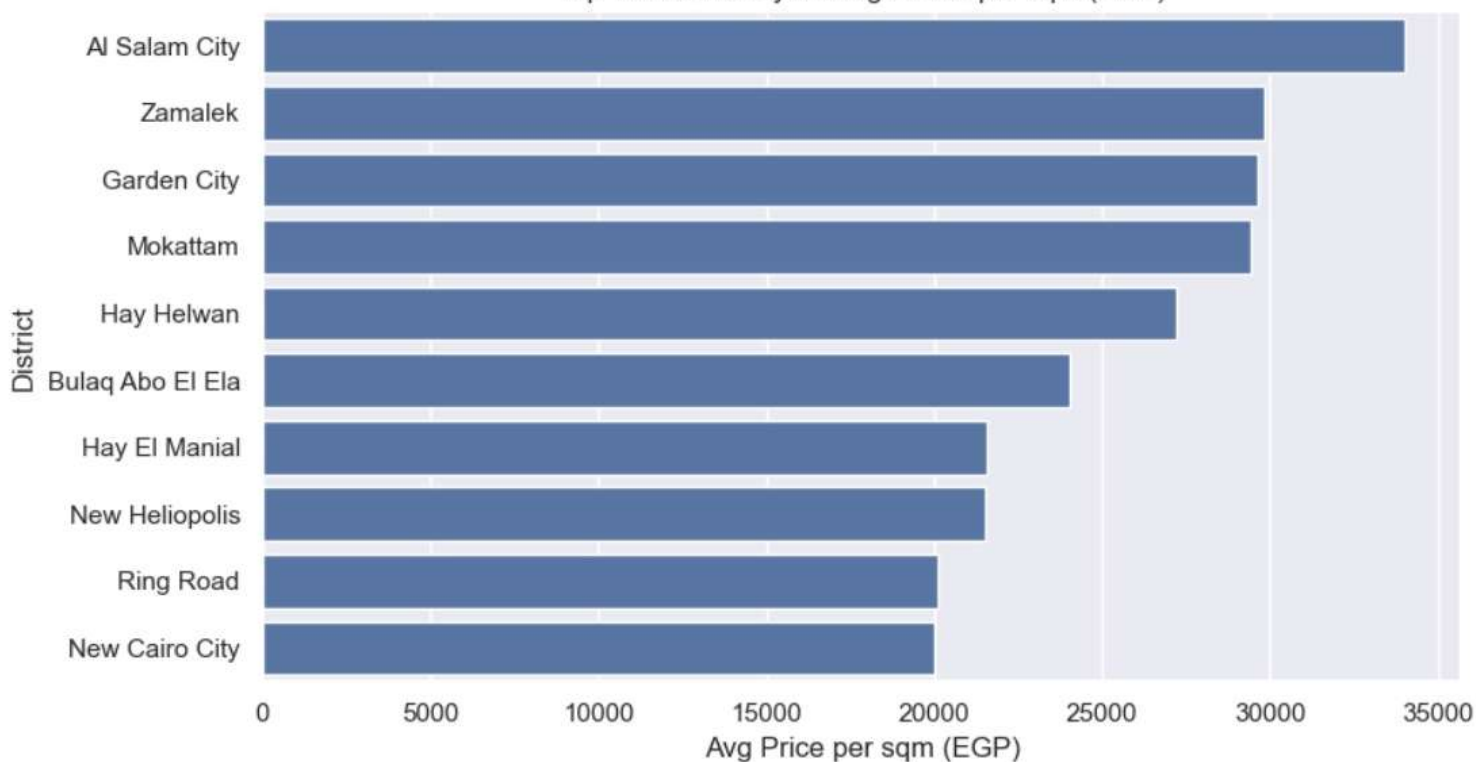
Column	Type	What It Tells Us
type	Categorical	Property category apartment, villa, duplex or more
title	Text	The listing headline (not used in analysis)
location	Text	Compound, district, and city (comma-separated)
bedroom	Integer	Number of bedrooms
bathroom	Integer	Number of bathrooms
size_sqm	Numeric	Property size in square meters
price	String / Integer	Listed price in EGP, or Ask If unknown

CLEANING THE DATA

- Removed 'Ask' in prices column: 412 rows had no actual price just 'Ask'. We dropped these since they can't be included in any price calculations.
- Fixed data types: The price column had commas in it (like 3,500,000) so we stripped those and converted everything into integers. Same for size and bedroom/bathroom columns some had formatting issues that needed fixing.
- Deduplicated rows: A small number of exact duplicate rows were removed.
- Removed outliers: I used IQR-based fencing on both size_sqm and price. Anything outside the $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ range was dropped. This got rid of extreme luxury properties and likely data errors.
- Parsed location: The location column was one messy string with compound, district, and city all squished together. We divided it into three clean separate columns

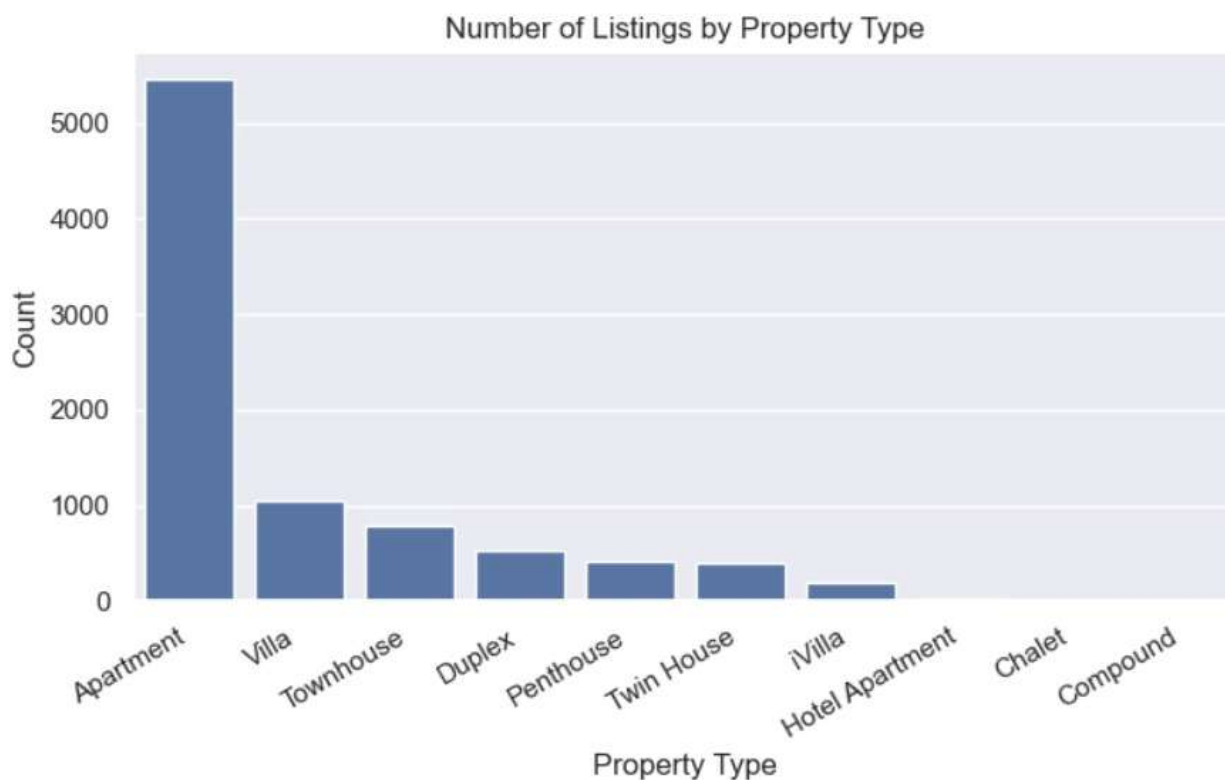
ANALYSIS

Top 10 Districts by Average Price per sqm (EGP)

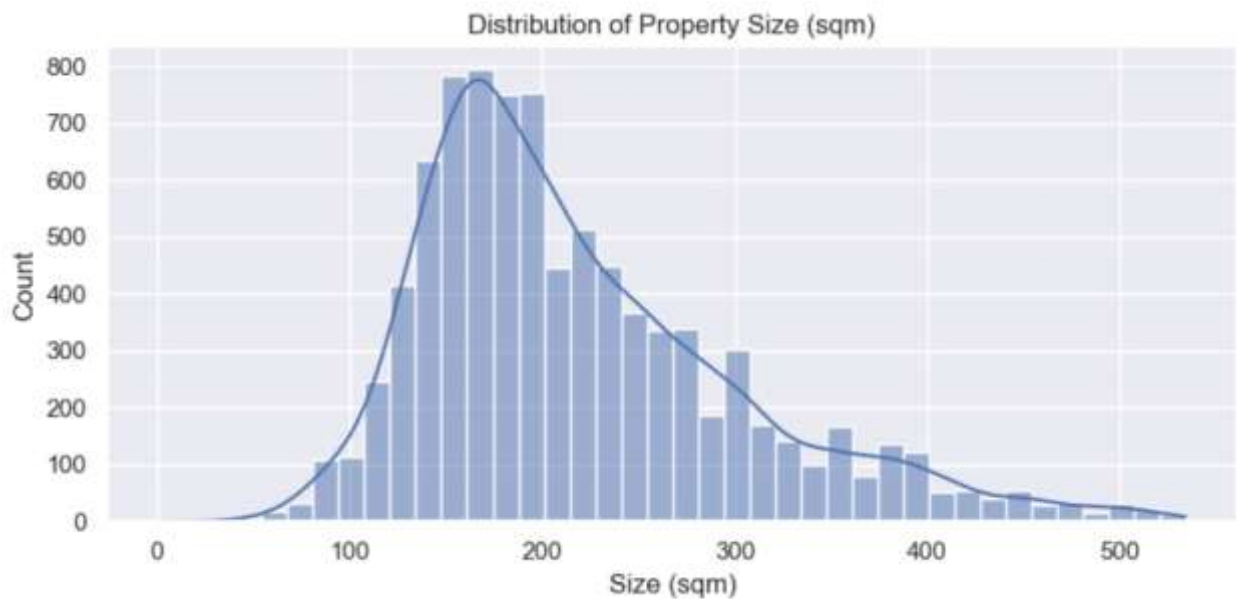


It highlights how powerful location is when it comes to property prices in Cairo. Al Salam City comes out on top, with an average price of over 34,000 EGP per square meter surprisingly even higher than well-known upscale areas like Zamalek and Garden City.

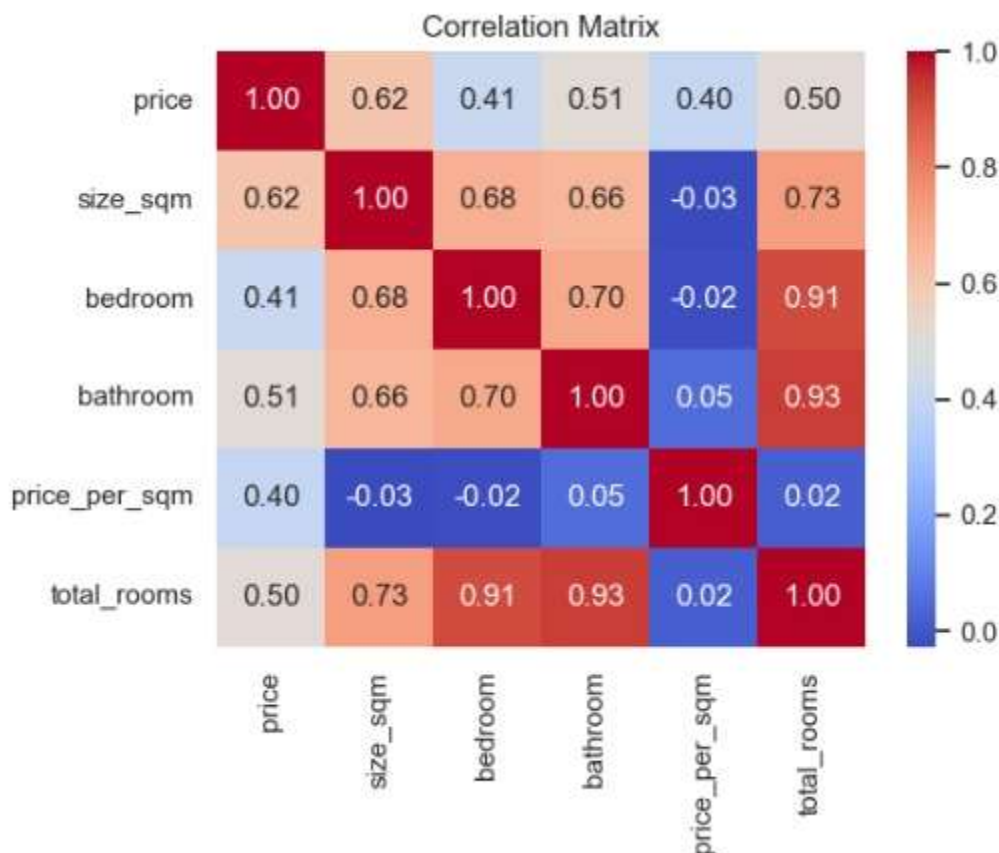
On the other end, New Cairo City appears at the bottom of this top ten list, averaging around 20,000 EGP. That gap is significant and shows just how much prices can vary depending on the area.



The bar chart shows the distribution of property types in the dataset. Apartments dominate the market by a huge margin, with over 5,000 listings, far exceeding all other categories. Villas come next with around 1,000 listings, while the numbers then gradually drop for townhouses, duplexes, penthouses, and twin houses. The remaining types like iVillas, hotel apartments, chalets, and compounds appear in very small numbers, indicating they are more niche or premium property options.

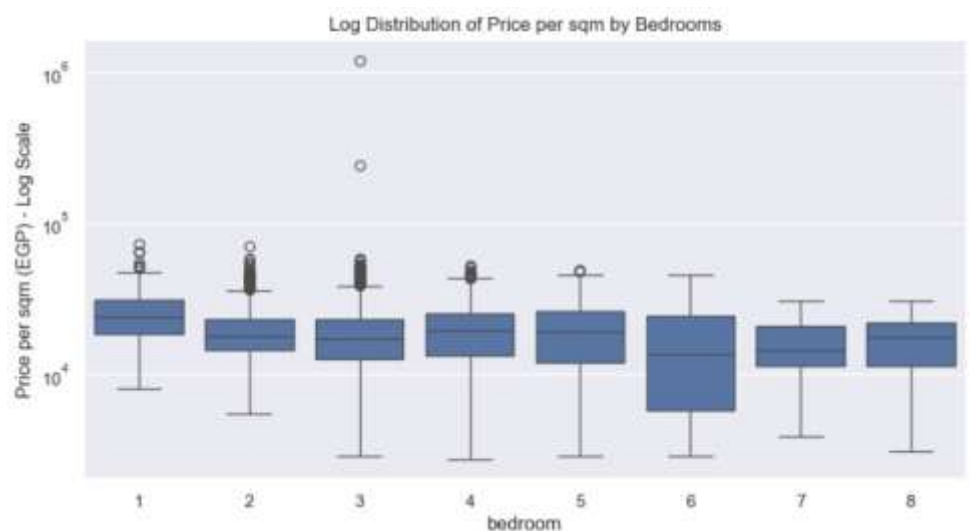


The histogram of property sizes shows a right-skewed distribution, meaning most listings fall in the smaller to mid-range sizes. Most properties are between 100 and 250 sqm, with the most common size around 150–175 sqm. As size increases beyond 300 sqm, the number of listings drops steadily, but a small tail extends past 500 sqm, showing that large luxury properties exist but are much less common



The heatmap basically shows how different features are connected. Deep red areas mean a strong relationship for example, things like total rooms, bedrooms, and bathrooms are very closely linked, with correlations going as high as 0.93, which makes sense since bigger homes usually have more of everything.

On the other hand, price_per_sqm doesn't seem to follow that pattern at all. Its row is mostly neutral (blue), which means it doesn't really depend on the size of the property or the number of rooms it behaves more independently.

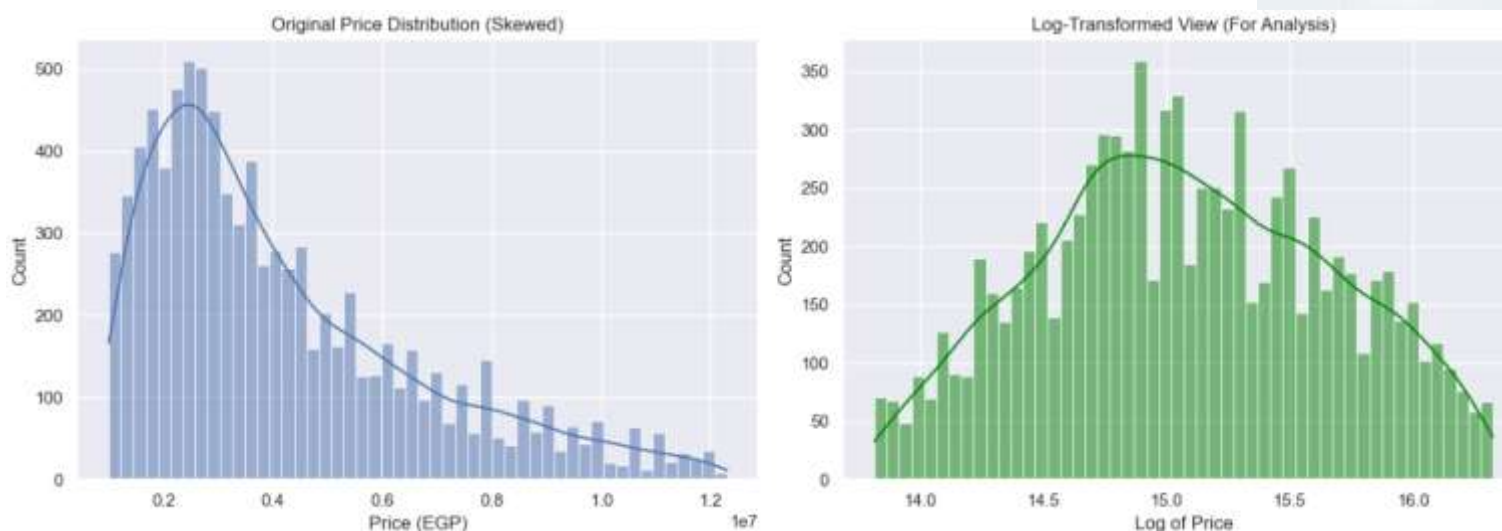


The visual compares property price per square meter across different bedroom counts using two scales. In the linear plot, extreme outliers stand out clearly especially a 3-bedroom property with a very high price but most of the data gets squeezed into a narrow band, making it hard to interpret. In contrast, the logarithmic scale spreads the data out more evenly, making the overall pattern clearer. It shows that median prices stay fairly consistent across 1 to 8 bedrooms, and the variation between groups looks much more balanced since the outliers are less dominant.



The scatter plot shows the relationship between property size and price, split by property type and using a log scale for price to handle the wide range of values. Overall, there's a clear trend that larger properties tend to be more expensive. Apartments mostly cluster in the smaller size range (under 250 sqm), while villas and twin houses appear on the larger end, often going beyond 300 sqm and reaching the highest prices.

There's also noticeable overlap between types like townhouses, duplexes, and penthouses in the mid-to-large size range, which suggests that once properties get bigger, the type and luxury level start to matter almost as much as size when it comes to pricing.



The two charts show how the data changes after normalization.

On the left, the original price distribution is very skewed to the right. Most properties are priced at the lower end around 2 million EGP, while a small number of expensive properties stretch the range all the way up to about 12 million.

On the right, after applying a logarithmic transformation, the data looks much more balanced. The extremely high values are compressed, and the distribution becomes closer to a normal (bell-shaped) curve.

This step is important because it makes the data easier to work with in statistical models. It reduces the impact of outliers and helps algorithms treat changes in price more consistently, rather than being overly influenced by a few very high values.

TESTING

To understand how price per square meter changes, we looked at both property type and district using two different statistical tests.

First, we used **ANOVA (F-test)**. This test helps compare the average prices across many groups at the same time. Since we're dealing with multiple property types and many districts not just two groups ANOVA is a good fit.

However, housing prices are usually not perfectly distributed. They tend to be skewed, which means ANOVA's assumptions might not always hold. That's why we also used the **Kruskal-Wallis test**. This is a non-parametric alternative that doesn't rely on normal distribution. Instead of comparing averages, it compares rankings, making it more reliable for this kind of data.

By using both tests together, we made sure our results are stronger and more trustworthy when checking how property type and district affect price per square meter.

HYPOTHESIS

What We Set Out to Test

H₀ Null Hypothesis

Location and property type have NO significant effect on price per sqm.

Any observed differences are due to random variation, not systematic factors.

H₁ Alternative Hypothesis

At least one factor: district or property type significantly predicts price per sqm.

Location and type create statistically distinct price groups that go beyond random chance.

Significance level: $\alpha = 0.05$ - Both parametric (ANOVA) and non-parametric (Kruskal-Wallis) tests applied for robustness

CONCLUSION

The statistical analysis of 8,851 cleaned property listings across 28 Cairo districts provides strong evidence that both location and property type are significant drivers of price per square meter. All four tests conducted two ANOVA and two Kruskal-Wallis rejected the null hypothesis at a significance level of 0.05, confirming that these factors are not merely coincidental but systematic predictors of pricing. Location emerged as the single strongest driver, with Al Salam City leading at approximately 34,000 EGP/sqm and a gap of nearly 70% separating top districts from average ones like New Cairo. Property type proved equally influential, with villas and twin houses commanding more than double the price of a typical apartment a premium that persists even when controlling for size. Notably, size alone was found to be a weak predictor, underscoring that buyers and investors who rely solely on square footage when evaluating Cairo properties are likely to make misinformed decisions. These findings collectively suggest that any pricing model, investment strategy, or urban policy targeting Cairo's real estate market must treat location and property type as primary variables rather than secondary considerations.